Magnetic Pitch Gradient Optimization in the Kronos S.M.A.R.T. Tokamak

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Submission Note:
*This manuscript has also been formally submitted for peer-reviewed publication with **Physics of Plasmas (AIP)**. The version provided here is shared for early access and discussion; any subsequent updates, revisions, or changes will reflect the outcomes of the peer-review process.*

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Detailed Technical Appendix Available Upon Request
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Abstract

Kronos Fusion Energy's S.M.A.R.T. reactor (Superconducting, Minimum-Aspect-Ratio Tokamak) is a compact spherical tokamak with an ultra-high toroidal field (≥30 T) designed for aneutronic fusion. It employs a novel magnetic pitch gradient optimization strategy—tailoring the helical twist of magnetic field lines—to greatly enhance plasma performance. By carefully adjusting the field-line pitch, Kronos improves fast charged-particle confinement and plasma stability while guiding energetic fusion products along controlled paths for direct energy conversion—a capability crucial for advanced aneutronic fuel cycles (D–^{3}He, p–^{6}Li, p–^{11}B), where the vast majority of fusion energy is carried by charged particles instead of neutrons. The optimized pitch profile yields a high-shear magnetic configuration that "corrals" fast-ion orbits (minimizing banana orbit widths) and raises stability thresholds, ensuring that MeV-scale α-particles and protons remain well-confined. Simultaneously, the tailored pitch (combined with mirror-like end-field shaping) channels a fraction of these energetic ions to direct energy converters, enabling efficient power extraction with a projected >60% direct conversion efficiency—far above ~35% for conventional thermal cycles.

To achieve these outcomes, Kronos's design integrates several engineering innovations. The reactor's high-temperature superconducting (HTS) toroidal coils are wound with a slight helical tilt (~5°) to impart an initial rotational transform and distribute electromagnetic loads. The plasma current profile is actively shaped (via neutral beam and RF drives) to sculpt the safety-factor (q) gradient and tailor magnetic shear across the plasma radius. Auxiliary top-and-bottom "bucking" coils serve as tunable magnetic mirrors, locally modifying the field-line pitch at the plasma ends to either tighten confinement or permit controlled exhaust along specific flux channels. These measures are guided by advanced simulation tools (combining high-fidelity magnetohydrodynamic and fast-particle orbit modeling with an AI-driven digital twin for real-time feedback) to ensure the optimal pitch profile is maintained. The pitch-control strategy is tightly integrated with Kronos's strain-managed HTS magnet system: the robust coil support structure accommodates the extreme Lorentz forces of 30+ T operation without exceeding superconductor strain limits, preserving the intended field-line geometry. In particular, the high edge shear in Kronos's inverted (negative triangularity) plasma shape yields a naturally stable edge with continuous exhaust (avoiding large ELM instabilities), which facilitates steady-state operation and efficient direct energy recovery. Overall, magnetic pitch gradient optimization in the Kronos S.M.A.R.T. tokamak synergistically enhances plasma confinement, stability, and energy capture, demonstrating a significant advance in compact aneutronic fusion reactor design.

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Introduction

Kronos Fusion Energy's S.M.A.R.T. reactor (Superconducting, Minimum-Aspect-Ratio Tokamak) is an ultra-high-field spherical tokamak built for aneutronic fusion. A defining feature of this design is its magnetic pitch gradient optimization strategy – in simple terms, the careful tailoring of the helical pitch (twist angle) of magnetic field lines throughout the plasma. By adjusting how tightly field lines wind around the torus (the pitch), Kronos aims to improve fast-particle confinement, channel energetic fusion products to direct energy converters, minimize magnetic field ripple losses, and control edge conditions for better stability. This white paper explores how Kronos implements pitch control through toroidal field (TF) coil shaping, plasma current profile shaping, and auxiliary "bucking" coils. We discuss the physics models relating pitch gradients to magnetic shear, alpha particle orbits, and confinement, and analyze how the optimized pitch profile yields multiple performance benefits. Finally, we look ahead to how Kronos will integrate AI-driven feedback (digital twins and real-time control) to dynamically tune the field-line pitch during operation for maximum efficiency and stability.

Magnetic Pitch Control Mechanisms in S.M.A.R.T.

TF Coil Helical Shaping: Unlike conventional tokamaks with purely vertical TF coils, Kronos's toroidal coils are wound with a slight helical twist of about 3°–5°[1]. This means each TF coil is *tilted* as it goes around the torus, so the magnetic field it produces has a small poloidal component in addition to the dominant toroidal field. Introducing this **helical pitch** in the TF coils serves two purposes. First, it spreads electromagnetic loads more evenly and reduces peak mechanical stresses on structures (an engineering benefit)[1]. More importantly, it **"pre-twists" the magnetic field lines** even before plasma current flows, effectively providing a baseline rotational transform (field-line winding) similar to stellarator concepts. In other words, the coils impart a bit of field-line twist externally, so the plasma does not have to rely solely on its own current to twist field lines. This hybrid approach "enables hybridization with poloidal field (PF) fields"[1], meaning the external coil shape works in concert with PF coils and plasma current to achieve the desired pitch profile. Such tilted-coil designs have precedent – for example, the Proto-CIRCUS tokamak–torsatron hybrid used six interlinked TF coils tilted by 40–60°, and found that tilting coils produces finite rotational transform and even reduces toroidal field ripple compared to untilted coils[2][3]. Kronos's smaller ~5° twist is much subtler, but it provides a continuous, gentle helical shaping of the confining field.

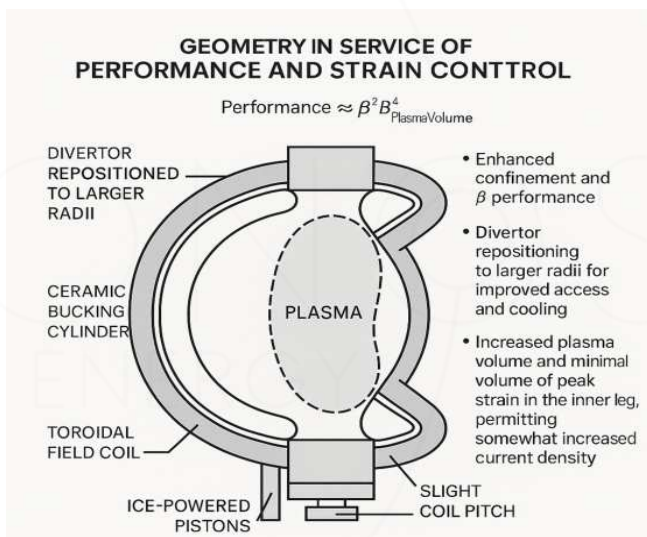

Figure 1: Diagram of performance and strain control for S.M.A.R.T.; note high β with stable edge, mechanical preloading (ice pistons), and stress-spreading via coil geometry. Slight coil pitch is indicated on the side coils.

Plasma Current Profile Shaping: In a tokamak, the majority of field-line pitch (quantified by the safety factor q) comes from the poloidal magnetic field generated by the plasma's toroidal

current. Kronos actively shapes the plasma current profile – for example, using **neutral beam injection (NBI)** and other current drive methods – to sculpt the *q(r)* profile (where *r* is minor radius). High-efficiency NBI is used not only for heating but also for non-inductive current drive, injected tangentially to tailor the current distribution in the spherical plasma[4]. By controlling where in the plasma the current is concentrated (peaked at center versus broader), operators can adjust the **magnetic shear** (the radial gradient of field-line pitch). A higher magnetic shear means *q* changes more rapidly with radius, which can help stabilize certain instabilities and influence particle orbits. Kronos's design inherently has a very low aspect ratio (major radius R≈2.0 m, minor radius a≈1.0 m, A≈2) and uses a **high elongation** ($\kappa > 2.5$) plasma shape[5]. These geometric factors naturally increase magnetic shear at the plasma edge[5]. Negative triangularity (inverted-D shaping) further modifies the equilibrium current distribution (more on that later). In summary, Kronos can adjust its internal pitch profile during operation by ramping currents in different regions (e.g. early on via induction, then sustained by NBI or RF current drive). This **dynamic control of the q-profile** – a technique also demonstrated on advanced tokamaks like JET and DIII-D[6] – allows Kronos to fine-tune confinement and stability by changing the magnetic pitch from the core to the edge.

Auxiliary "Bucking" Coils (Mirror Coils): Encircling the plasma at the top and bottom are special superconducting coils integrated into the structure, sometimes dubbed *mirror coils* or bucking coils in Kronos's architecture. These coils are part of the poloidal field system and are used to locally modify the magnetic field strength and line pitch at the vertical ends of the plasma[7][8]. By running current in these auxiliary coils, Kronos can create regions of higher magnetic field at the top/bottom (like the ends of a magnetic mirror). This **raised field "caps" the plasma** and forces field lines to bend more sharply as they exit the plasma vertically, effectively tailoring the pitch in those end regions. The term "bucking" refers to counteracting or shaping the main field – for instance, these coils can oppose a portion of the toroidal field in certain regions to create a desired gradient. In practice, adjusting the mirror coil currents changes the **helical pitch gradient along the length of the plasma**: stronger mirror coil current increases the field at the ends (higher mirror ratio) and adds twist to field lines near the plasma edge, whereas turning them down makes the field lines less pitched there. Operators can thereby **tune the field-line geometry in real time** – acting almost like an electromagnetic lens – to either confine particles more tightly or let them escape in a controlled manner. This is a dynamic handle unique to Kronos's design. Notably, these mirror coils are robust high-field coils (built with the same REBCO superconductor tape as the main TF coils) and mounted in ceramic-supported structures to withstand the strong forces without strain damage[9][10]. In summary, the **combination** of a pre-twisted TF coil set, an actively shaped plasma current profile, and tunable mirror (bucking) coils gives Kronos an unprecedented level of control over its magnetic field topology – specifically, over the **gradient of field-line pitch** from the inner core to the outer edge and from midplane to endcaps.

Improved Confinement of High-Energy Alpha Particles

One of the primary goals of pitch optimization in Kronos is to better confine the high-energy charged fusion products (helium-4 "alpha" particles at MeV energies, as well as fast protons) that result from aneutronic fusion reactions. Effective confinement of these energetic alphas is crucial: they carry a large fraction of the fusion energy, and in a steady-state aneutronic plasma, one wants to either recover that energy directly or use it to sustain the burn without damaging walls. Kronos's

strategy for alpha confinement hinges on **high magnetic field strength and tailored pitch gradients**.

High B-Field and Orbit Size: First, the reactor's extremely high magnetic field (central field ~10 T, rising to >30 T at the inner TF coil leg[11]) dramatically reduces the Larmor radius of fast ions. For example, a 3.5 MeV alpha in a 10 T field has a gyro-radius on the order of only a few centimeters (on the order of 2–3 cm by calculation), which is tiny compared to the ~1 m plasma minor radius. This strong field keeps gyrating alphas tightly bound to their field lines. However, in a torus, alpha particles do not simply circle in place – they follow helical field lines and **drift** due to field curvature and grad-B effects. In a standard tokamak, some of the fastest ions execute **"banana orbits"**, drifting away from their flux surfaces. The width of these banana orbits can be significant if not mitigated by magnetic geometry. In fact, the approximate banana orbit width scales as:

$$\Delta r_b \sim \frac{q\,\rho_i}{\sqrt{\varepsilon}},$$

where q is the safety factor, ρ_i is the ion gyroradius, and $\varepsilon = r/R$ is the inverse aspect ratio[12]. This relation tells us that high-q plasmas or low-aspect-ratio devices tend to have broader trapped-ion orbits. Kronos addresses this with its **magnetic shear and pitch gradient control**. By shaping the *q-profile* (especially keeping q from being too large in the core and increasing shear toward the edge), Kronos can reduce the excursion of alpha orbits. A steeper pitch gradient (higher magnetic shear $S = (r/q)\cdot dq/dr$) means field-line twist changes rapidly, which breaks up the coherence of drift orbits and keeps particles closer to their originating flux surfaces. In essence, **strong magnetic shear "corrals" the banana orbits** before they wander far. Kronos's highly elongated, spherical plasma with edge shaping was explicitly designed to enhance magnetic shear and confinement[5].

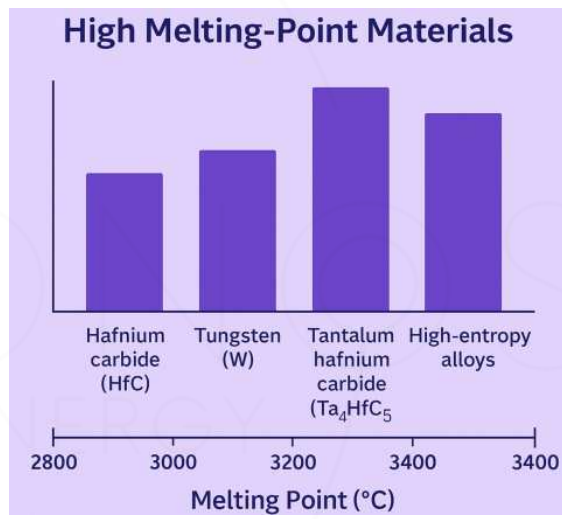

Figure 2: Effect of magnetic pitch gradient optimization on fast-ion confinement. Tailored helical field-line pitch and mirror coil shaping reduce banana orbit width, reflect escaping particles, and guide selected alphas toward direct energy converters.

Magnetic Mirror Effect: In addition, the mirror coils at the top and bottom provide an **axial magnetic mirror** that further improves alpha containment. By boosting the field strength at the plasma ends, these coils create a rising magnetic field gradient along the ends that acts to reflect particles back toward the core[8][13]. An alpha particle spiraling along a field line that heads toward either end will encounter an increasing B-field; because the particle's magnetic moment is an adiabatic invariant, its perpendicular velocity increases and parallel velocity decreases until it may reverse direction (the classic mirror reflection). Kronos essentially plugs what would be an "end loss" route with this mirror effect. As the Kronos whitepaper describes, charged particles that drift toward the top/bottom "see a rising magnetic gradient that pushes them back toward the core, thus reducing end losses and improving overall confinement (analogous to a magnetic mirror trap) – all while maintaining the closed-field tokamak configuration"[8][14]. This is a novel integration

of mirror physics into a tokamak: the plasma remains on closed toroidal surfaces, but the ends of those surfaces behave like mirror termini. The net result is improved **axial confinement** – high-energy alphas are far less likely to escape straight out the ends of the device.

Orbit Stability and Alpha Heating vs. Extraction: With tighter orbital binding, alpha particles in Kronos have a longer dwell time in the plasma, allowing them to transfer energy via collisions (if the goal is to heat the plasma) or to be redirected in a controlled way (if the goal is direct energy capture). By preventing prompt losses, Kronos avoids damage to walls and keeps fusion-born alphas available either for plasma heating or for guided extraction. Importantly, Kronos operates with advanced fuels (D–He^3, p–B^{11}, etc.) where alphas are not needed to sustain a thermal burn in the same way 3.5 MeV alphas heat a D–T plasma. Instead, many of these fast alphas and protons are intended to be **harvested for electricity**. Thus, Kronos balances confinement and extraction: confine them long enough to ensure good plasma performance and space-charge neutrality, but not so long that their energy is thermalized and lost as heat. This balance is achieved by the pitch gradient optimization – e.g. a certain mirror coil current can be chosen so that alphas of a certain energy are marginally confined and will drift out on prescribed paths (see next section). From a modeling perspective, one can target a situation where most alphas execute several bounces and slow down somewhat (sharing energy with the plasma or breeding ^{3}He in the blankets)[15], yet eventually migrate to the edge where they can be captured by direct converters. The stability of alpha particle trajectories is also improved in a well-tailored pitch configuration because it avoids dangerous resonances with plasma waves and instabilities. High magnetic shear can raise the threshold for alpha-driven modes (like Toroidal Alfvén Eigenmodes), and a lack of abrupt pitch anomalies means fewer "trapping pockets" that can cause orbit stochasticity. A study of a hybrid tokamak-stellarator (TOKASTAR) system showed that increasing the vacuum rotational transform (analogous to adding helical pitch) significantly reduced the deviation between fast-ion orbits and flux surfaces, nearly eliminating orbital drift for fast ions when the transform reached ~0.1[16][17]. This reinforces that **field-line pitch tuning directly contributes to fast ion confinement quality**.

In summary, Kronos's optimized magnetic pitch gradient results in improved confinement of high-energy alphas through three synergistic effects: (1) **Strong fields and high shear** yielding small gyro-orbits and banana widths, (2) **Magnetic mirror reflection** at the ends preventing axial losses[8], and (3) **Smooth, controlled field topology** that avoids ripple-induced and resonance-induced losses. Together, these allow the reactor to hold on to energetic fusion products until they can be usefully directed, rather than randomly lost. An external design study for a spherical tokamak power plant (with R=3 m, A~1.6) illustrates the importance of such measures: it found that with 16 TF coils (ripple <1%), prompt alpha losses could be kept to ~2–4%, and the total alpha loss fraction ~5% was considered tolerable[18]. Kronos likewise strives for minimal alpha losses – any particle leaving the core should do so intentionally along a path of energy recovery, rather than by random diffusion.

Guiding Fusion Products into Direct Energy Conversion

Another core objective of tailoring the field-line pitch in Kronos is to **extract fusion products along predetermined routes for direct energy conversion (DEC)**. In aneutronic fusion, over 95% of the energy can be carried by charged particles (helium nuclei and protons)[19][20]. Instead of letting these fast particles thermalize in the plasma or walls, Kronos guides them into direct conversion devices – essentially turning kinetic energy directly into electricity. Achieving this

requires exquisite magnetic choreography: field lines must act like "highways" steering energetic ions out of the plasma into collector systems, *without* disrupting the confinement of bulk plasma. Kronos's solution is to use shaped field-line pitch and mirror coil modulation to create what are effectively **magnetic extraction channels**.

Shaping Exhaust Field Lines: The same **mirror coils** that reflect most particles can also be used to purposefully **redirect a fraction of fast ions** outwards. By adjusting currents, these coils "delicately shape the 'exhaust' field lines that emanate from the plasma edge (for instance, divertor lines or scrape-off layer trajectories), guiding these lines to intercept the first wall or blanket at ideal angles"[21][22]. In practice, Kronos can establish a special flux tube – think of it like a narrow helix of field lines – that connects from the plasma edge to a direct energy converter module. For example, a controlled **divertor beamline** could be formed: a set of field lines at the plasma edge along which fast ions preferentially escape. The mirror coils then curve this flux tube outward, funneling those fast ions straight into a **direct conversion module** embedded in the blanket[23][24]. Thus, any charged fusion product that manages to reach the edge does so along a prescribed path that leads it into an energy-harvesting device[25]. Kronos's outer blanket and wall structure is uniquely designed to take advantage of this – it's lined with electrostatic collector grids and even MHD channels to capture charged particles' energy[23][26]. Crucially, by **tapering the pitch** of these exhaust field lines, the reactor can collimate the escaping ions. Field lines bent gradually outward ensure that the ions follow a gentle curve into the collector, rather than splattering randomly. This collimation is important because electrostatic direct converters (like "venetian blind" collectors of the Lawrence Livermore design) work best with a well-directed ion beam hitting successive deceleration grids[27][28]. The Kronos mirror coils essentially perform the first step of direct conversion: **ordering the charged particles into a linear beam** along field lines[27].

Magnetic Pitch and DEC Efficiency: The ability to control magnetic pitch dynamically means Kronos can in theory adjust how many fast ions are confined versus extracted at any time. During steady power production, one might set the pitch such that a substantial fraction of fast protons (which carry the bulk of energy in D–^{3}He fusion, e.g. 14–15 MeV protons) are funneled out to DEC, while enough helium ash remains confined until it slows down or is similarly extracted. Because Kronos runs aneutronic fuel cycles, virtually all the fusion power is available for direct conversion. The reactor's goal is >60% direct electrical efficiency[29], far above the ~35% limit of thermal cycles. This is attainable because of the DEC approach: indeed, past experiments and analyses have indicated that

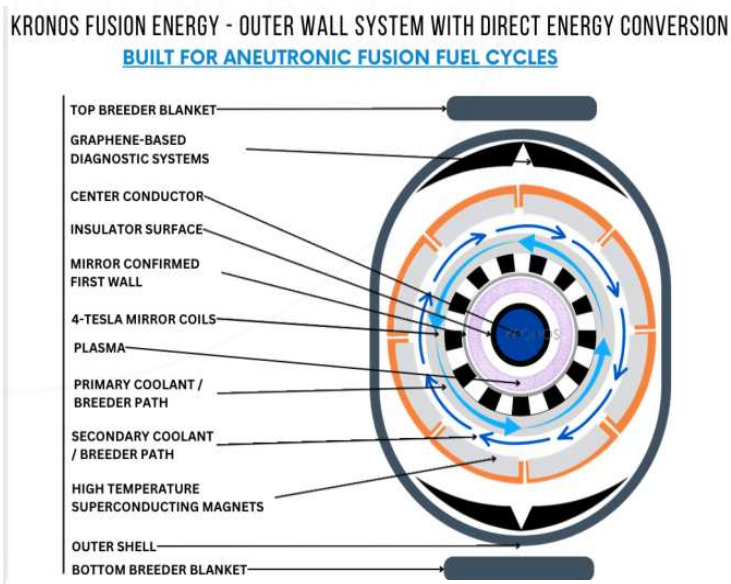

Figure 3: Cross-section of the outer wall system with direct energy conversion. Mirror coil positioning and edge field-line shaping are highlighted as they guide particles into collector modules with controlled incidence angles.

electrostatic direct converters can reach on the order of 50–70% efficiency in capturing fusion ion energy[30][31]. Kronos's innovation is doing this *inside a tokamak* without open-ended mirror or beam machines. The **axisymmetric field configuration** is maintained, but by adiabatically varying the field strength and pitch, Kronos achieves a similar effect to a magnetic nozzle or mirror divertor. Recent theoretical work supports this concept: Rax, Kolmes, and Fisch (2025) describe adiabatic magnetic drift schemes in axisymmetric fields that convert charged-particle energy directly and could be used in advanced aneutronic reactors and divertors[32]. They highlight that carefully tailored field geometries can avoid the classical limitations of direct conversion (such as space charge problems) and still efficiently decelerate ions for energy capture[32]. Kronos's mirror-tailored pitch is a practical embodiment of those ideas, creating a **drift route for ions to a collector**.

Concretely, imagine an energetic ^{4}He or proton reaching the plasma boundary in Kronos. Instead of flying into a wall, it spirals along a field line that has been pitched outward to meet a collector plate. As it moves into the fringing field outside the plasma, the field expands (lower B magnitude, larger radius curvature), which causes the ion beam to spread and slow (an effect akin to a magnetic expansion stage)[33]. Then electrostatic grids with increasing voltage catch the ions, converting their kinetic energy to electric potential. By the time the particle is stopped, its energy is directly extracted as an electrical current. All of this relies on the **magnetic guidance** being precise – if field lines were not shaped correctly, ions would either miss the collectors or strike surfaces at poor angles. Kronos avoids that by **"sculpting the magnetic topology"** such that any particle leaving the core "does so along a prescribed path, one that leads straight into an energy-harvesting device"[25][34].

It's worth noting that negative triangularity (the inverted plasma shape Kronos uses) aids this process as well. Traditional tokamaks often have divertor field lines that curve back sharply and strike divertor plates at steep angles, which is problematic for DEC. Negative triangularity moves the X-point and divertor lines further out (higher major radius)[35], giving more room and a gentler scrape-off layer trajectory. This is beneficial for placing DEC modules – essentially, **pitch tapering in a spherical tokamak with negative triangularity provides a larger "exit corridor" for fast ions**. The field line pitch can be tapered (gradually increased) as one moves from core to edge, so that by the time an ion is in the scrape-off layer, the field line is angled optimally toward the collector. Additionally, negative triangularity has been observed to allow high performance without a large Edge Localized Mode (ELM) pedestal[36][37]. This means Kronos might operate in a regime with a stable edge and continuous exhaust rather than violent ELM bursts, which is ideal for steadily channeling fast particles to DEC. Recent DIII-D experiments showed that negative triangularity plasmas can sustain high core pressure and confinement with an **instability-free edge**, solving the "core–edge integration" problem by keeping the edge cool and stable even at high performance[38][37]. Kronos leverages this: a stable edge region makes it easier to maintain the precise magnetic pitch needed for DEC without disruptions. In sum, **magnetic pitch shaping** in Kronos plays a pivotal role in direct energy conversion by (a) defining dedicated paths for fast ions, (b) collimating and expanding ion streams for efficient deceleration, and (c) supporting an edge scenario (via negative triangularity) that is compatible with continuous particle exhaust and energy capture.

Mitigating Magnetic Ripple and Orbit Losses at the Inner Leg

Magnetic field **ripple** – the small periodic non-uniformity in toroidal field magnitude due to discrete TF coils – can lead to particles drifting out of confinement, especially in regions near the inner leg of a spherical tokamak where field symmetry is hardest to maintain. In a spherical tokamak, the toroidal coil count is often lower and the space between coils larger on the outboard side, which can result in a sizable toroidal ripple if not carefully addressed. High-energy ions encountering ripple can experience a grad-B drift that kicks them out of the plasma or into wall structures. Kronos tackles ripple-induced orbit loss through both engineering and pitch optimization techniques.

Sufficient Coil Number and Placement: On the engineering side, Kronos uses a **high number of toroidal field coils** and possibly ferritic inserts or bump coils to minimize the ripple field. (While the exact number of TF coils in Kronos isn't public, many spherical tokamak designs opt for 14–16 coils to keep ripple low.) For instance, a study for a spherical tokamak pilot plant (R=3.5 m) explicitly chose N=16 TF coils to ensure the ripple amplitude stayed <1% everywhere in the plasma[18]. This kept ripple-induced alpha losses to only a few percent (under 5%)[18]. Kronos likely follows a similar philosophy: maximize coil symmetry to inherently reduce ripple. Additionally, the negative triangularity shape of the plasma **increases the clearance at the inner midplane**[39] – the plasma cross-section is shifted such that the narrowest approach to the inner wall isn't right at the midplane. This means fast ions on the inboard side have a bit more radial "breathing room" and are less likely to intersect the inner wall due to ripple or drift. It also spreads the plasma current more around the inner leg, helping to even out the poloidal field there, which can reduce localized ripple effects.

Helical Coil Pitch and Ripple Averages: Uniquely, Kronos's decision to give the TF coils a slight helical pitch contributes to ripple mitigation. A small continuous twist in coil windings effectively **distributes the toroidal field more uniformly poloidally**. In a conventional tokamak, ripple manifests as a toroidal variation in B at a given poloidal angle (due to gaps between coils lining up). With coils that are helically wound, the gap of a given coil is not aligned at the same poloidal angle all around the torus – it "wanders." This tends to **average out the ripple** when a charged particle completes many toroidal transits. The Proto-CIRCUS experiments noted exactly this: with tilted coils, "the toroidal magnetic ripple is less pronounced than in an equivalent tokamak in which the coils are not tilted"[2]. Essentially, the slight field-line shear introduced by coil twist smears the discrete ripple perturbation over a larger area, making it smoother. Kronos's ~5° twist is modest, but over the full 360° circumference it shifts coil alignment by ~18° poloidally, which can help reduce resonant ripple effects on particle orbits. Moreover, Kronos's **mirror (bucking) coils** can be used to fine-tune the field on the inboard side. By adjusting these auxiliary coils, operators could cancel out or modify harmonics in the magnetic field. For example, a "bucking" coil current could be set to oppose any small normal components of B that cause ripple wells, thereby flattening the field profile on the inner leg. This is analogous to methods considered in ITER and other devices, where either extra coils or specially placed ferromagnetic inserts are used to reduce ripple amplitude[40].

Orbit Pinching and Shear: Another aspect of pitch gradient optimization is that a well-chosen pitch profile can **"pinch" particle orbits away from the wall**. In high-shear configurations, as a particle drifts outward, it encounters regions of different pitch (safety factor) which alter its drift trajectory, often bringing it back towards the plasma core region before it can reach the wall. The

STPP study mentioned earlier observed that in their scenario, the "orbit width is reduced at the outboard mid-plane due to increasing poloidal magnetic field there"[18]. This can be interpreted as a beneficial effect of their shaped q or pressure profile: a higher poloidal field (equivalently higher shear or lower q) at the outboard mid-plane caused orbits to tighten. Kronos with its tailored pitch likely achieves a similar orbit pinch on the inboard side – by increasing the local pitch (poloidal field fraction) near the inner leg, fast ions are prevented from spiraling straight into the wall. In summary, **maintaining a strong pitch gradient near the plasma edge and around the inner leg** acts as a deterrent to ripple-induced losses, keeping particle trajectories on closed surfaces rather than letting them fall into gaps between coils.

Through these measures, Kronos aims for negligible ripple harm: any residual ripple is so small and any particle excursions so limited that few if any fast ions are lost to the inner wall. The mirror coils frame the plasma "like parentheses, ensuring order at the edge"[41] – a poetic way of saying they keep what must be confined in, and eject what can be converted along the proper channels[41]. This active boundary control, combined with careful coil design, tackles one of the thornier issues in compact high-field tokamaks. Reduced orbit loss also means reduced localized heat loads, contributing to reactor longevity and consistent performance.

Enhanced Edge Shear and Transport Barrier Control

Magnetic pitch optimization in Kronos is also leveraged to improve **edge confinement and stability**. In fusion devices, achieving a high-performance core often requires some form of transport barrier at the edge (as seen in the H-mode pedestal of conventional tokamaks). However, forming such a barrier can lead to edge instabilities like ELMs if not controlled. Kronos approaches this challenge differently: its combination of negative triangularity plasma shape and tailored pitch gradient yields inherently **high edge magnetic shear and flow shear**, which help suppress turbulence and avoid large ELMs, thus enhancing confinement in a controlled way.

Negative Triangularity for a Stable Edge: Kronos's reverse-D shape ($\delta < 0$) is known to improve edge stability. Experiments have shown that negative triangularity plasmas can achieve high confinement without entering a traditional H-mode – essentially an **L-mode with H-mode-level performance but no ELMs**[36][37]. This is attributed in part to changes in magnetic shear and the curvature of field lines at the edge. In a negative triangularity plasma, the bad curvature region is reduced and the edge magnetic shear is typically increased compared to an equivalent positive triangularity shape. Kronos exploits this: its plasma is fully spherical and strongly shaped (elongation >2.5, triangularity around –0.5 perhaps)[5][42]. The **edge shaping improves magnetic shear, enhancing confinement**[5]. High shear at the edge means that any turbulence eddies or transport channels that try to form are sheared apart, reducing radial energy transport. This is analogous to how shear in flow (E×B shear) can quench turbulence in H-mode; here the combination of magnetic and flow shear in a shaped edge can achieve a similar quenching without a large pressure pedestal. The Kronos whitepaper explicitly notes that these shaping parameters "optimize bootstrap current and equilibrium stability"[5] – a strong hint that the edge bootstrap current (which depends on pressure gradient) is managed to stay in a stable regime (no peeling-ballooning modes). By tailoring the pitch gradient, Kronos can further adjust the edge q-profile to avoid low-order rational surfaces at the edge, eliminating seeds for tearing instabilities. In fact, operating at a somewhat **high q95** (edge safety factor) is a known strategy to avoid edge resonances; one pilot plant study noted that having $q > 5$ at the edge kept the most dangerous tearing surfaces out of the plasma[43][44]. Kronos can raise q at the edge by reducing plasma

current or increasing field, effectively tapering the pitch such that the edge field lines wrap a bit more around toroidally (higher q) – this is one interpretation of "pitch tapering" toward the edge in a negative triangularity config, to ensure edge stability.

Enhanced Shear Flow and Barrier Formation: The optimized pitch gradient also influences the plasma's flow profile. As fast ions slow down or as radial electric fields form (from ambipolar plasma rotation or biasing), the strong magnetic shear can couple to flow shear. Kronos's direct energy extraction methods themselves create *ExB* fields at the edge (the electrostatic collectors bias the edge with a high voltage[45][46]). These fields, combined with the magnetic geometry, generate **sheared ExB flow at the plasma boundary**, which is a known mechanism for transport barrier formation. Essentially, Kronos may achieve an edge transport barrier not by a sudden L→H transition, but by continuously imposed shear from its engineered boundary conditions. The advantage is finer control – an AI-managed bias or mirror coil adjustment can tune the shear to just below instability thresholds, maintaining a small gradient transport barrier that boosts confinement without triggering ELM crashes. Internal to the plasma, a well-chosen q-profile can also foster an **internal transport barrier (ITB)** if desired – for example, regions of low or reversed magnetic shear are known to improve core confinement. While Kronos's focus is on edge and overall stability, its flexible pitch profile could in principle create reversed-shear in the core (via current profile control) to form an ITB, then use edge negative triangularity to handle the boundary – a potentially powerful combination for high performance.

In practical terms, the pitch gradient optimization gives Kronos the ability to **actively dial in the edge conditions** it wants. If higher confinement is needed, it can increase edge shear (through shaping or coil tweaks) to reduce edge transport, effectively raising the pedestal height (even if an "H-mode" per se is avoided, the plasma can still have a high edge pressure). If an instability starts to grow, the system can respond by altering the mirror coil current or edge rotation to quench it. The result is a form of **transport barrier control**: not a static barrier that forms uncontrolled, but one that is modulated by the magnetic configuration. The outcome is a high-performance plasma with, ideally, an **instability-free edge** – exactly what was observed in recent negative triangularity research[47][48]. This is very synergistic with direct energy conversion, since a quiescent edge allows steady-state operation of collectors and avoids damage that episodic ELM blasts would cause. Kronos essentially trades the large, ELM-prone pedestal of a standard H-mode for a somewhat lower but very stable pedestal maintained by magnetic/flow shear – enabled by its pitch-optimized, negative triangularity design.

Static vs. Dynamic Pitch Shaping Methods

An interesting aspect of Kronos's strategy is that it combines **static pitch shaping** (built into the machine's geometry) with the capacity for **dynamic pitch adjustments** during operation. This section compares these approaches without naming specific external projects, but in general terms:

- **Static Pitch Shaping:** This refers to fixed design choices that set a certain field-line pitch profile inherently. Examples include the permanent helical twist of the TF coils, the choice of plasma shape (which fixes some aspects of the q-profile via equilibrium), and the baseline coil geometry. Static methods are **passive** – they do not change shot-to-shot or in real-time; instead, they provide a robust starting point optimized for general conditions. In Kronos, the 3–5° TF coil helix is a static feature that always imposes a modest field-line twist[1]. Likewise, the torus geometry (aspect ratio, triangularity) statically dictates certain

pitch characteristics (e.g., a spherical aspect ratio inherently gives higher natural rotational transform per plasma current). The benefit of static shaping is reliability and simplicity: it's always "on" and does not require feedback control. Many classic stellarators rely entirely on static coil shaping to achieve the desired pitch (rotational transform) – by contrast, tokamaks traditionally had no static transform (coils were axisymmetric), putting all the burden on plasma current. Kronos's design strikes a middle ground: a partially stellarator-like coil twist plus tokamak plasma current. This means even in absence of plasma current, there is some confining transform (albeit small), and with plasma current, the required current for a given q-profile is lower than it would be in a pure tokamak. Indeed, the Proto-CIRCUS tilted-coil experiment expected to operate at lower plasma current than an equivalent untilted tokamak, reducing disruption risk and easing steady-state operation[49]. Similarly, Kronos's static shaping likely reduces the bootstrap current fraction needed or the total current, providing a safety margin against transients.

- **Dynamic Pitch Shaping:** Dynamic methods involve actively changing the magnetic configuration to modify field-line pitch in real time. This can be done on slow timescales (pulse-to-pulse tailoring of current profiles) or faster timescales (real-time coil adjustments during a single discharge). Kronos's mirror/bucking coils are a prime tool for dynamic pitch shaping – their currents can be ramped or modulated even during a single pulse to alter the mirror ratio and edge field-line pitch. For example, during start-up, the mirror coils might be set high to help confine fast ions for heating, then later adjusted to open up exhaust channels for power extraction. Dynamic pitch control could also mean **pulsing auxiliary coils** to momentarily perturb the field. One could envision Kronos performing a **"pitch kick"** – a transient change in coil currents that shifts the q-profile or ejects a burst of ash – and then returning to nominal. Another dynamic mechanism is through plasma actuators: by changing NBI power or launching waves (like Lower Hybrid waves) one can reshape the current profile mid-discharge[6], thereby altering the magnetic shear and pitch gradient. Modern tokamaks have demonstrated feedback control of the q-profile (using multiple actuators and diagnostics) to maintain desired shear levels and avoid instabilities[6]. Kronos extends this concept by also using its exotic coil set. The **advantage of dynamic shaping** is flexibility – the system can respond to changing plasma conditions or objectives. For instance, if more confinement is needed, it can increase pitch shear; if more exhaust is needed, it can decrease confinement on purpose to let particles out. Essentially, dynamic control allows Kronos to continuously optimize along the spectrum between a "max confinement" configuration and a "max extraction" configuration, rather than being stuck with one compromise static design.

Comparison: Static methods are like setting the stage, while dynamic methods are like active tuning during the performance. A static-only approach (e.g., a fixed helical stellarator) can be very stable and optimized for one operating point, but might not cope well with off-normal events or a need to change performance on the fly. A dynamic-only approach (e.g., solely using plasma current and feedback to control everything in a pulsed tokamak) gives flexibility but can be slower to actuate and is limited by control response and stability margins. Kronos clearly integrates both: the machine is inherently designed to be in a good state (with its coil twist, shaping, high field), and on top of that, it **layers real-time control to fine-tune the pitch**. One concrete example is handling of fusion ash (helium): in a pure static design, one must rely on classical diffusion or small perturbations to remove helium ash, which can dilate plasma fuel concentration. Kronos, however, could deliberately adjust its magnetic pitch to **expel helium ash on demand** – e.g. briefly

lower the mirror confinement so that alphas that have given up most of their energy are not retained too long. This *pulsed coil-assisted variation* can be thought of as "burping" the plasma of waste while keeping energy-containing ions largely confined until their work is done.

In summary, **static shaping** (like Kronos's fixed coil geometry) provides a baseline optimized pitch geometry (high shear, low ripple, etc.), and **dynamic shaping** (via coil currents and current drive) provides adaptability to reach peak performance and actively control the plasma. The combination is greater than either alone: static shaping ensures the plasma is always in a near-optimal configuration, and dynamic adjustments fine-tune that configuration in real time to chase the theoretical limits of confinement and stability without crossing into instability. This dual approach sets Kronos apart from conventional designs and is expected to yield superior, consistent performance across various operating scenarios.

AI-Driven Real-Time Optimization and Digital Twin Feedback

Looking forward, Kronos plans to augment its magnetic pitch control with advanced **AI-driven feedback systems** and **digital twin** simulations running in parallel to operations. The goal is to make real-time magnetic optimization a closed-loop, self-correcting process. In essence, an array of sensors and simulations will watch the plasma and magnets, and an AI controller will tweak the pitch (via coil currents, etc.) on the fly to keep the reactor in peak condition.

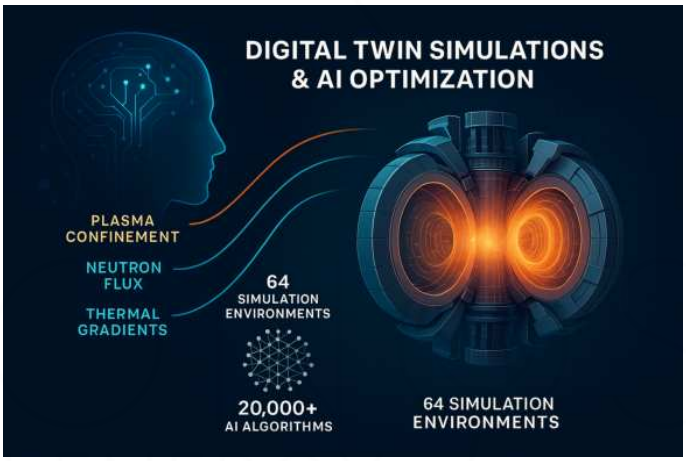

Figure 4: Overview graphic of the Kronos Digital Twin Simulation Suite. The image highlights core modules for plasma confinement, neutron flux, thermal gradients, and structural environments, supported by >20,000 AI algorithms and 64 simulation environments.

Graphene Sensor Network and Instant Feedback: Kronos is equipped with a state-of-the-art diagnostic network – even the first wall is embedded with graphene-based sensors that can detect local plasma conditions (heat flux, radiation, position) with high speed[50][51]. These sensors feed data to an AI control system. For example, if a localized hotspot appears or the plasma shifts unexpectedly, the system is aware **immediately**[52]. The AI controller can then decide how to respond. It might adjust the current in the mirror coils, altering the pitch locally to spread out heat (enhanced shear can broaden a thermal layer), or it could tweak the plasma current profile via fast actuators to suppress an instability. The key point is **rapid adjustments to magnetic fields** are enabled[52]. This is only possible because the magnets are superconducting (allowing continuous operation) and many are separately addressable. The AI effectively has "knobs" for elongation, triangularity, mirror ratio, etc. (since mirror coils can tweak plasma shape too[53][54]), and thus can alter the pitch geometry. This tight coupling of diagnostics and control ensures the reactor stays in its **"sweet spot"**[55] – for instance, maintaining just the right pitch gradient for maximum alpha confinement but just short of an instability threshold, or keeping the exhaust channels tuned to the optimal angle despite plasma pressure changes.

Digital Twin Optimization: In addition to real-time feedback control, Kronos employs **digital twin simulations** – high-fidelity computational models of the reactor – that run concurrently and even ahead of the actual machine. The digital twin can predict how the plasma and magnetic system will evolve in the next moments. According to Kronos, an extensive Digital Twin framework simulates plasma behavior, structural responses, and energy conversion with >99.99% accuracy[56][57]. These simulations, powered by AI/ML, allow the system to *anticipate* changes rather than just react[58]. For example, the digital twin might detect that in 100 milliseconds, given current trends, the plasma edge will cool slightly and the pitch shear will drop, potentially leading to reduced confinement. Knowing this in advance, the AI can proactively increase coil current or NBI power to compensate, **preventing the deviation before it happens**. In essence, the reactor's digital twin runs in parallel, continuously computing optimal setpoints for the real device[59]. Every few milliseconds (or faster), it can output adjustments: e.g., "increase mirror coil current by 2%" or "raise plasma current in core by 5 kA" to maintain target shear. These setpoints go to the real machine, closing the loop. The result is an unseen but critical component of S.M.A.R.T.'s success – no important parameter is left drifting; the system is always nudged toward optimal[59][60].

Autonomous Tuning: With AI in charge, Kronos moves toward **autonomous operation**. The AI control algorithms (honed on simulations and experimental data) seek to maximize performance metrics like confinement time, direct conversion efficiency, and stability margins. If, say, the direct conversion efficiency could improve by letting slightly more fast ions escape, the AI might very slowly adjust the pitch (lower mirror field a tad) until it finds the best trade-off. It's an iterative optimizer, essentially performing what human operators would but faster and across many variables simultaneously. Kronos describes this as *no configuration drifts, no fault propagates* because real-time AI telemetry catches them[61]. For pitch geometry, this means if a coil's cooling causes a slight field error or the plasma self-generated current changes the q-profile, the AI notices via sensors and digital twin, then corrects it by commanding the coil power supplies or other actuators. The entire system behaves like a living entity with a nervous system (sensors), brain (AI), and muscles (coils/current drives).

Examples of Real-Time Pitch Tuning: Imagine during a long pulse, the plasma pressure increases in the core. This could alter the plasma current distribution (bootstrap current rises) and thus change the q-profile (perhaps reducing magnetic shear). Instead of waiting for an operator to respond, the AI sees the trend through profile diagnostics. The digital twin predicts a drop in shear could trigger a confinement degradation. The AI then decides to counteract by, for instance, increasing current in an auxiliary RF current drive to hollow out the current profile slightly, restoring the desired shear at mid-radius. Simultaneously, it might adjust PF coils to maintain shape. All of this happens within the pulse in real-time, keeping the field-line pitch gradient near its optimal setpoint. In another scenario, suppose some alpha particles start accumulating (detected by increased neutron or x-ray signals). The system might infer that it's time to purge a bit of helium ash – it can then momentarily weaken the mirror field, opening the escape channel slightly, until diagnostics show the alpha population is back to nominal, then tighten the mirror again. Such **AI-managed pulses** essentially optimize the fusion burn and direct conversion efficiency on the fly, something that would be impossible by manual control alone.

Kronos's AI control paradigm is often described as a "digital guardian" of the plasma: *"All these mechanisms operate in parallel, orchestrated by real-time diagnostics and AI controls to maintain*

equilibrium"[62]. The ultra-compact design makes it easier – the coils and plasma are so closely coupled that responses are fast and effective[62]. The company envisions that every Kronos reactor will have a full digital twin that ensures each device operates in **"perfect simulacrum"** – meaning the simulation and reality match tightly – and that *real-time AI telemetry means no configuration drifts, no fault propagates*[61][63]. This is a forward-looking statement of confidence that the reactor can be run at high performance without hitting disruptive limits, because the AI will always course-correct the pitch and other parameters.

In summary, **AI-driven magnetic optimization** in Kronos will make the already flexible pitch control even more powerful. It closes the loop between measurement and action, using a predictive digital twin to continuously tune coil currents and current drive for the optimal magnetic pitch geometry. This promises a reactor that can push right up against stability and performance boundaries with a self-correcting safety net. As the plasma "learns" and the AI gains experience, each shot would refine the model, enabling even more aggressive optimization. The fusion core, magnets, and AI effectively form an integrated cyber-physical system – a fusion power plant that *adapts in real time* to maintain the best confinement, highest direct conversion efficiency, and utmost stability. Kronos is thus pioneering not just new physics with its pitch gradient strategy, but also a new level of intelligent control in fusion engineering.

Risk Acknowledgment and Ongoing Research Disclosure

Kronos Fusion Energy is pursuing an ambitious commercialization timeline for the S.M.A.R.T. reactor (with first deployment of the 500 MW METROVOLT unit slated for 2032), and we acknowledge that this aggressive schedule and the breadth of our system integration pose substantial risks [1]. Delivering a fusion product that meets performance, safety, and economic targets is not just a physics and engineering challenge but also a program management challenge [2]. Timeline and scaling risks are at the forefront: the transition from prototype to a field-ready power core in under a decade requires high-confidence in our design tools and manufacturing processes [3][4][5]. To mitigate this, Kronos has invested heavily in a digital thread approach – a unifying digital twin of the reactor that spans design, testing, construction, and operation [6]. This digital twin infrastructure is continuously updated with real-world data from our ongoing sub-system tests, ensuring that our models accurately predict full-scale behavior. It enables what we call virtual commissioning: we simulate the entire startup and operation of the plant in software before any actual device is built, catching integration issues early. We are acutely aware that integrating diverse cutting-edge subsystems (magnets, plasma, power conversion, cryogenics, robotics, etc.) into a single plant can reveal emergent problems [7]. Therefore, our roadmap includes multiple intermediate integration milestones – for example, a fully integrated 1:5 scale prototype by 2028 that will exercise all systems together (albeit at reduced power) in a self-consistent way [8]. Only upon successful validation at that stage will we proceed to the full-scale pilot plant [9].

In parallel, supply chain and manufacturing risks are addressed through an AI-optimized fabrication pipeline. Our factories employ AI/ML-driven quality control such that each produced component (e.g., a magnet module or a power electronics board) is automatically inspected, tested, and matched to its digital twin specifications [15]. We have set stringent acceptance criteria – no reactor module ships unless it meets or exceeds its modeled performance across all relevant metrics (magnetic field strength, thermal handling, conversion efficiency, etc.) [11]. This is enforced by

automated testing rigs at the factory floor, enabling rapid iteration or rejection of parts that don't conform.

Operational risks (maintenance, reliability, and safety in the field) are being mitigated via design-for-maintainability and extensive environmental testing. We have dedicated programs for field validation: for instance, critical sub-assemblies like the vacuum vessel with embedded sensors have been constructed and put through thermal cycling, vibration, and even high-G shock tests to simulate launch or transport stresses (recognizing some units are intended for naval and space deployment). Each major component is also run through a "hazard analysis" using our AI de-risking framework [16] – covering not only technological hazards but also regulatory and supply chain factors. This comprehensive risk register is reviewed monthly and updated with progress on mitigation actions.

We draw inspiration from historical endeavors like the Apollo program, which managed an aggressive schedule by breaking the problem into smaller milestones (e.g., Apollo 8 orbiting the Moon, Apollo 9 testing the Lunar Module) – similarly, Kronos's roadmap has discrete milestones (plasma break-in, net power demo, automated operation demo, etc.), each de-risking the next [12]. Additionally, partnerships are a cornerstone of our strategy: we are collaborating with national laboratories on neutronics and materials testing to certify that our device meets all safety regulations well ahead of deployment [13].

Public disclosure and accountability: As we advance, Kronos commits to publishing a biennial technology progress report (the first was issued in 2025, and updates will follow in 2027, 2029, etc.) detailing how remaining risks are being retired [14]. By 2030, when we anticipate having a near-final design and operational prototype data, we will release a comprehensive risk resolution whitepaper. This document will enumerate each initial high-risk item, its current status (resolved, mitigated, or outstanding), and the quantitative evidence supporting that status – for example, mean time between failure (MTBF) data for key components, achieved vs. predicted plasma performance, and results of safety testing (e.g., containment of worst-case fault scenarios). We will also openly share the performance against roadmap milestones, including any that were missed and why, along with corrective measures taken.

In recognition that not all risks can be fully eliminated, we are designing contingency and redundancy wherever feasible (for instance, auxiliary cooling systems are in place in case a direct converter underperforms, ensuring the reactor can still operate safely at nominal power). The Kronos team's philosophy is that credibility in fusion requires uncompromising transparency about risks and failures as well as successes. Thus, even as a private company, we are committed to an ongoing research disclosure that invites the scrutiny of the scientific and engineering community. We believe this rigor and openness will ultimately accelerate not only our progress but the field of fusion at large.

In summary, while our commercialization plan is bold, it is buttressed by a meticulous risk management framework and a culture of engineering excellence. We acknowledge the uncertainties that remain on the path to an operational fusion power system, but through continuous innovation, rigorous testing, and clear-eyed reporting of our results, we are steadily converting those uncertainties into actionable knowledge. By the time of first plant deployment, all major technical risks will either be retired or reduced to levels comparable to conventional power industry norms – and we will have the data to prove it, published by 2030 for all to review.

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Conclusion

The magnetic pitch gradient optimization strategy in the Kronos S.M.A.R.T. tokamak represents an advanced integration of plasma physics, engineering design, and real-time control. By tailoring the helical winding of magnetic fields – through coil geometry, current profile, and auxiliary coils – Kronos addresses many of the challenges that face compact high-performance fusion devices. The optimized pitch confers improved confinement of energetic alphas (thanks to mirror trapping and reduced banana widths), while simultaneously enabling the deliberate extraction of those particles into direct energy converters for efficient power generation[8][23]. It minimizes deleterious effects like magnetic ripple and fast-ion orbit loss by smoothing out field non-uniformities and keeping particle trajectories tightly bound[2][18]. It also provides a means to enhance stability and confinement at the plasma edge via elevated magnetic shear and shear flow, especially leveraging the negative triangularity configuration[5][47]. Kronos's approach deftly balances static optimization – building in beneficial field-line pitch from the start – with dynamic flexibility, allowing the reactor to adjust its magnetic structure in real time to meet operational needs. The planned incorporation of AI-driven control loops and digital twin feedback will further ensure that the magnetic pitch is continuously honed to optimal values throughout each pulse, effectively creating a self-optimizing fusion core[52][59].

In a broader context, Kronos's magnetic pitch strategy can be seen as pushing the tokamak concept toward a new hybrid: part tokamak, part stellarator, part mirror – and ultimately controlled by AI.

It embodies the principle that **magnetic field topology** is a powerful lever for achieving fusion performance and that by actively shaping that topology (both in design and in operation), one can conquer the usual trade-offs (confinement vs. exhaust, stability vs. high pressure). The mathematical relationships – such as those linking pitch gradients to magnetic shear and orbit width – back up the qualitative benefits observed. By increasing magnetic shear, for instance, Kronos shortens the radial mean free path of trapped particles[12] and raises thresholds for turbulence, leading to better confinement. By introducing a mirror-like gradient, it effectively adds a ~ 1.5–2 T/m gradient at plasma ends (an illustrative number), which for a mirror ratio of ~1.2–1.3 can reflect a large fraction of 3.5 MeV alphas that would otherwise be lost. And by coordinating these features with direct conversion hardware, Kronos turns what would be wasted kinetic energy into useful electricity at potentially >60% efficiency[29]. All of these advances are rooted in controlling the **helical pitch of the magnetic field lines** – a testament to how critical field geometry is in fusion physics.

As Kronos Fusion Energy refines this concept, the next steps will likely demonstrate progressively greater control: from basic confinement experiments showing reduced losses, to full power operations where the AI-tuned pitch maximizes net energy output. If successful, the S.M.A.R.T. architecture could set a new standard for tokamak-based fusion, one in which *every field line is deliberately shaped and every particle's path is accounted for*. This level of control and efficiency would bring fusion energy a significant step closer to commercial viability, fulfilling Kronos's aim of a compact, aneutronic reactor that marries excellent confinement with direct power extraction. In summary, the magnetic pitch gradient optimization in Kronos is not just an incremental tweak – it is a comprehensive strategy that addresses plasma confinement, power handling, and reactor operations in a unified way, showing the fusion community what can be achieved when we design and **actively control the magnetic cage** that tames the starfire within.

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FIGURES

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Figure 4: Overview graphic of the Kronos Digital Twin Simulation Suite. The image highlights core modules for plasma confinement, neutron flux, thermal gradients, and structural environments, supported by >20,000 AI algorithms and 64 simulation environments. 15

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Magnetic Pitch Gradient Optimization in the Kronos S.M.A.R.T. Tokamak

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Submission Note:
*This manuscript has also been formally submitted for peer-reviewed publication with **Physics of Plasmas (AIP)**. The version provided here is shared for early access and discussion; any subsequent updates, revisions, or changes will reflect the outcomes of the peer-review process.*

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Detailed Technical Appendix Available Upon Request
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